III. Analogy.

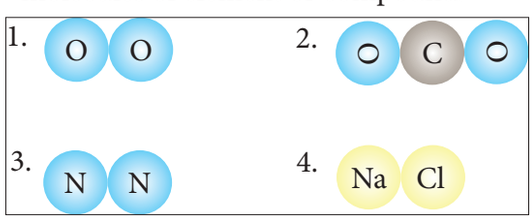
1. Mercury : Liquid at room temperature :: Oxygen: _____.
2. Non-metal conducting electricity : _____ :: Metal conducting electricity : Copper
3. Elements : Combine to form compounds :: Compounds :_____.
4. Atoms : Fundamental particle of an element :: _____ : Fundamental particles of a compound.

IV. State true of false. If false, give the correct statement.

1. Two different elements may have similar atoms.
2. Compounds and elements are pure substances.
3. Atoms cannot exist alone. They can only exist as groups called molecules.
4. NaCl represents one molecule of sodium chloride.
5. Argon is mono atomic gas.

V. Answer in brief.

1. Write the chemical formula and name the elements present in the following compounds.
   a. Sodium chloride
   b. Potassium hydroxide
   c. Carbon dioxide
   d. Calcium oxide
   e. Sulphur dioxide
2. Classify the following molecules as the molecules of element or compound.



3. What do you understand by chemical formula of a compound? What is its significance?
4. Define the following terms with an example for each.
   a. Element
   b. Compound
   c. Metal
   d. Non-metal
   e. Metalloid
5. Write the symbols for the following elements and classify them as solid, liquid and gas.
   Aluminum, Carbon, Chlorine, Mercury, Hydrogen and Helium
6. Classify the following as metals, non-metals and metalloids.
   Sodium, Bismuth, Silver, Nitrogen, Silicon, Carbon, Chlorine, Iron, Copper
7. Classify the following as elements and compounds.
   Water, Common salt, Sugar, Carbon dioxide, Iodine and Lithium
8. Write the chemical formula for the following elements.
   a. Hydrogen b. Nitrogen
   c. Ozone d. Sulphur
9. What are elements? What are they made of? Give two examples.
10. Define molecule.
11. What are compounds? Give two examples.
12. Give an example for the elements derived from their Latin names.
13. What is atomicity of elements?
14. Calculate the atomicity of H_2SO_4.

VI. Answer in detail.

1. Differentiate metals and non-metals.
2. Explain the characteristics of compounds
3. Describe the different ways in which we can write the symbols of elements. Give appropriate examples.

VII Std Science Term-1 EM Unit 3.indd 39 23-01-2025 12.33.19 PM